

Liang-Ching Tao

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Research Interests

Learning-guided planning and sequential decision-making; reinforcement learning; tree search with learned policy and value guidance (MCTS/PUCT); planning with learned models, abstractions, and hierarchy.

Education

National Chung Hsing University (NCHU), Taiwan — M.S. in Computer Science and Engineering 2026

Advisor: Prof. Pi-Chung Wang. Graduate GPA: 3.84/4.00. Thesis: *A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks*.

National Chengchi University (NCCU), Taiwan — B.S. in Computer Science 2023

Research Experience

Learning-Guided Multi-Charger Planning with Neural-Guided Search Sep. 2026–present

- Extending LP-BTS to multi-charger scheduling: graph-based policy representations, canonical actions, action-space reduction, and neural-guided search over the enlarged joint action space.

Learning-Guided Planning in Large Dynamic Action Spaces: Budgeted Tree Search for One-to-Many Mobile Charging (LP-BTS) 2026–present

- Formulated one-to-many mobile charging as planning over a state-dependent action space of physical charging stops that is rebuilt from the live geometry at every decision and exceeds 10^3 actions.
- Built LP-BTS: a graph proposal policy concentrates a 32-arm candidate support, a learned value critic evaluates leaves, and edge-budgeted PUCT compares short simulated futures before an action is committed.
- Trained the proposal policy by imitation of a search expert followed by a teacher-free update toward its own search visit distributions; the value critic regresses terminal survival.
- Designed matched ablations that isolate proposal support from explicit search, and a prespecified, sealed 30-scenario confirmatory bank evaluated once. Preprint [P1].

HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling 2025–2026 — M.S. thesis, NCHU

- Formulated coupled charging, routing, and allocation as a structured sequential decision problem; developed a two-level architecture in which Q-learning selects regions or scheduling decisions and force-aware routing determines charger movement. Preprint [P2]; related conference presentations [C2] and [C5].

Preprints

- [P1] **Liang-Ching Tao**, Pi-Chung Wang. “Learning-Guided Planning in Large Dynamic Action Spaces: Budgeted Tree Search for One-to-Many Mobile Charging.” [arXiv:2609.17429](https://arxiv.org/abs/2609.17429) (cs.LG; cross-list cs.AI), Sep. 2026.
- [P2] **Liang-Ching Tao**, Pi-Chung Wang. “HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks.” [arXiv:2609.13901](https://arxiv.org/abs/2609.13901) (cs.NI; cross-list cs.LG), Sep. 2026.

Conference Papers and Presentations

First author on every item. MAMM 2026 is the 8th International Conference on Microactuators, Microsensors and Micromechanisms; the MAMM items are accepted for presentation and are not part of published proceedings. One-page abstracts are linked; full texts are available on request, and extended versions are being prepared for journal submission.

- [C1] **Liang-Ching Tao**, Wei-Cheng Tai, Pi-Chung Wang. “Adaptive Charging Threshold and Detour Scheduling for Wireless Rechargeable Sensor Networks via Deep Reinforcement Learning.” *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Accepted for oral presentation. [[abstract](#)]
- [C2] **Liang-Ching Tao**, Pi-Chung Wang. “HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks.” *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Accepted for poster presentation. Extended English version of [C5]; preprint [P2]. [[arXiv](#)]
- [C3] **Liang-Ching Tao**, Po-Cheng Fu, Pi-Chung Wang. “DTSC: A Dual-Tier Sector-Based Cooperative Mobile Charging Scheme for Wireless Rechargeable Sensor Networks.” *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026.

Accepted for poster presentation. [abstract]

- [C4] **Liang-Ching Tao**, Wei-Chen Lin, Pi-Chung Wang. “CHASE: A Corridor-Heuristic Charging Scheduler for Ground and Aerial Vehicles in Wireless Rechargeable Sensor Networks.” *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Accepted for poster presentation. [abstract]
- [C5] **Liang-Ching Tao**, Pi-Chung Wang. “A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks” (in Chinese). *2026 MC&WASN Conference*, National Chung Hsing University, Taiwan, Aug. 24–25, 2026. Presented as a poster. [PDF]

Other Research Experience

Research Assistant — Raman Spectral Analysis and Detection System, NCHU 2025–2026

- Built the full pipeline: spectral preprocessing, a 1D residual-convolutional encoder with multi-head attention, and binary Sudan Red I–IV detectors served through a FastAPI/Next.js application. [code]

Research Assistant — Citrus Crop Disease and Pest Recognition, NCHU 2025–2026

- Trained and compared EfficientNetV2-S and ResNet50 for class-imbalanced citrus disease and pest recognition; applied Focal Loss, MixUp/CutMix, and Grad-CAM; exported models to ONNX for inference. [code]

Teaching Experience

Teaching Assistant — Computer Networks (Undergraduate), NCHU 2025

Instructor: Prof. Pi-Chung Wang. Assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Teaching Assistant — Computer Networks (Professional Master’s Program), NCHU 2025

Instructor: Prof. Pi-Chung Wang. Assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Teaching Assistant — New Trends in Network Technology (Graduate Course), NCHU 2025

Instructor: Prof. Pi-Chung Wang. Assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Technical Skills & Graduate Coursework

Machine Learning & AI: Reinforcement Learning; Deep Learning; Graph-based Learning; Planning & Search (MCTS/PUCT).

Programming & Frameworks: Python; C/C++; PyTorch; NumPy; pandas; scikit-learn; FastAPI; Flask; Next.js; JavaScript/TypeScript.

Systems & Tools: Linux; Git/GitHub; CUDA; Docker; ONNX; Kafka; LaTeX.

Graduate Coursework — AI & Data: Deep Reinforcement Learning (A–); Advanced Artificial Intelligence (A–); Natural Languages Processing (A); Internet of Things (IoT) Data Analysis and Applications (A+).

Graduate Coursework — Systems & Hardware: Advanced VLSI Design (A+); Digital Testing (A); System-on-chip Testing (A–); New Trends in Network Technology (A–).